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FUNDAMENTALS

Multi-agent AI

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Abstract

Multi-agent artificial intelligence (MAAI) represents a foundational shift in the automation of knowledge work, moving beyond static workflows toward adaptive systems of interacting AI-based agents. These agents perceive, reason, and coordinate in real time to address complex, context-rich tasks that traditionally require human expertise. Drawing on the conceptual roots of process automation, agentic information systems, and AI, this paper introduces a structured, five-component framework that conceptualizes MAAI as a layered architecture composed of foundation model, data-centric perception and action, dynamic orchestration, agent-integrated workflow, and interaction interface. This framework disentangles the technical, organizational, and human-facing dimensions of MAAI, offering researchers and practitioners a systematic lens to analyze and design agent-based AI automation. The framework further structures three research pathways focused on advancing technical capabilities, enabling organizational integration, and addressing socio-technical implications such as fairness, accountability, and labor transformation. Together, these contributions establish a foundation for interdisciplinary inquiry into how MAAI reshapes work, coordination, and digital value creation.

Keywords Agentic information systems · Artificial intelligence · Process automation · Multi-agent systems

JEL classification: O3 · M15 · M21 · C45 · C63

Introduction

Imagine a scenario in which a customer contacts their insurance provider to inquire whether recent damage to their vehicle is covered under their current policy. The inquiry—submitted via a customer portal or mobile app—includes a short textual description and photographs of the damage. This seemingly simple request triggers a series of internal processing steps within the insurance company. A customer service representative must interpret the inquiry and extract relevant details. A policy analyst then needs to assess whether the reported case aligns with the customer's contract terms, potentially consulting internal databases or regulatory frameworks. Subsequently, a claims processor cross-checks the case with historical claims and pricing information to evaluate plausibility and prepare a response. Each of these roles contributes expert knowledge and judgment

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to tailor a case-specific answer. This is a representative scenario of complex knowledge work, which goes beyond merely processing information and instead requires the creation, distribution, and application of (expert) knowledge. Skilled individuals autonomously manage information and data to navigate complex situations and produce tailored outcomes (Drucker, 1999; Pyöriä, 2005; Schultze, 2000). This type of knowledge work requires a deep understanding of framed experience, values, contextual information, and expert insight. Accordingly, the tasks involved are typically unstructured, non-routine, and intricately tailored to specific situations (Davenport et al., 1996; Davenport & Prusak, 1998; Heerwagen et al., 2004; Janz et al., 1997).

With the advancement of artificial intelligence (AI), automation in information systems (IS) is entering a new phase. While earlier technical automation approaches such as robotic process automation (RPA) focused on deterministic, rule-based execution of routine tasks (Hofmann et al., 2020; Jimenez-Ramirez et al., 2019), recent developments increasingly extend into unstructured, context-sensitive domains (Acharya et al., 2025). The growing accessibility of AI-as-a-Service (AIaaS) (Lins et al., 2021) and low-code development platforms (Bock and Frank, 2021) lowers technical barriers, while generative AI models provide adaptive reasoning and language capabilities that were previously out of reach (Fetzer et al., 2025; Feuerriegel et al., 2024; Schneider et al., 2024).

In light of this convergence of increasingly capable AI models, accessible AI infrastructure, and lowered development barriers, we witness the emergence of a new phenomenon: **Multi-agent AI (MAAI)**. Unlike traditional automation pipelines, MAAI consists of ensembles of AI-based agents (autonomous entities capable of perception, decision-making, and action) whose interactions are not governed by predetermined workflows but emerge dynamically in response to context. Building on core concepts from agentic IS, such as agency, delegation, and governance, MAAI extends these ideas to multi-agent settings in which multiple AI-based agents must coordinate, delegate, and adapt their behavior collectively to address complex, unstructured tasks (Baird & Maruping, 2021; Zhou et al., 2024). This development does not emerge in a conceptual vacuum. Instead, MAAI revives long-standing principles of multi-agent systems (MAS) research in computer science (CS) (Ferber & Weiss, 1999; Wooldridge, 2009), which emphasized the intertwined challenges of agent and society design (da Silva & Demazeau, 2002; Demazeau, 1997), and integrates them with IS discourses on process automation (Engel et al., 2022; Liu et al., 2025), agentic IS (Baird & Maruping, 2021; Feuerriegel et al., 2024), and AI-enabled cognitive capabilities[1].

From this convergence arises both the promise and the ambiguity of MAAI. On the one hand, MAAI extends the MAS tradition into IS contexts, equipping digital agents with foundation model (FM)-based cognitive capabilities in socio-technical environments (Liu et al., 2025; Schneider et al., 2024). On the other hand, prior IS research has shown that introducing computational agents raises foundational definitional and organizational questions: what counts as an agent, how should collectives of agents be governed, and where do human actors fit within dynamically orchestrated workflows? (Baird & Maruping, 2021; Svanberg et al., 2024). These conceptual questions highlight the need for a conceptual framework that makes sense of MAAI as both a technological phenomenon and an IS construct.

To this end, we propose a five-component "onion-like" framework that structures MAAI along interdependent layered debate spaces: (1) *foundation model*, (2) *data-centric perception and action*, (3) *dynamic orchestration*, (4) *agent-integrated workflow*, and (5) *interaction interface*. These components act as layered conceptual containers that allow the IS community to analyze MAAI systematically, disentangle its complexity, and engage with it through shared theoretical foundations. By disentangling these layers, our MAAI framework not only explains *what MAAI is* and *how MAAI works*, but also provides the fundamentals needed to *systematically investigate MAAI further*. In doing so, our framework positions MAAI not as a fleeting industry trend under labels such as "agentic AI," but as a theoretically grounded paradigm that builds on established traditions while enabling new pathways for research in IS. As novel MAAI developments accelerate (Acharya et al., 2025), MAAI holds the potential not only to restructure knowledge work but also to redefine occupational roles and reconfigure digital markets.

In the following sections, we provide a structured exploration of the MAAI landscape. The "Conceptual roots" section examines the conceptual roots of MAAI, offering theoretical grounding that situates the phenomenon within existing streams of IS research. The "Multi-agent AI" section introduces our framework for MAAI, articulating five interrelated components that collectively define the core layers of AI-based MAS. The "Challenges and emerging pathways for the IS community in electronic markets" section then outlines key implementation challenges and identifies emerging research pathways, demonstrating how the proposed framework enables a systematic investigation of MAAI's technical capabilities, organizational and economic integration, and socio-technical implications. Together, these sections establish a foundational understanding of MAAI as

[1] Here, "cognitive capabilities" refers to the core mental functions of AI-based agents (learning, reasoning, memory, decision-making, and contextual understanding) as articulated in research on cognitive systems (Vernon et al. 2007).

both a conceptual innovation and a transformative force in the evolving field of knowledge work automation.

Conceptual roots

The conceptual foundations of MAAI originate in the long-standing tradition of MAS research in CS. MAS emerged as a response to problems that exceed the capabilities of monolithic systems, highlighting that many computational tasks require distributed problem solving, coordination, and collective intelligence (Ferber & Weiss, 1999; Wooldridge, 2009). Classic MAS theory identifies two inseparable design challenges: the *agent design* problem (how to construct autonomous entities capable of perception, reasoning, and action) and the *society design* problem (how to organize multiple agents into effective collectives that can pursue complex objectives in shared environments) (Wooldridge, 2009). These represent the micro- and macro-level perspectives of MAS, yet they are not orthogonal: individual design choices shape emergent group behaviors, while organizational structures impose constraints on agent reasoning and interaction. To systematize this interdependence, MAS research advanced integrative frameworks such as the vowel paradigm (da Silva & Demazeau, 2002; Demazeau, 1997), which characterizes MAS across five complementary views: agent (individual capabilities and decision logic), interaction (communication and coordination protocols), environment (the structuring role of shared contexts), and organization (roles, institutions, and coordination mechanisms).

Building on this foundation, the following subsections trace how these classical ideas resurface in MAAI. First, we distinguish MAAI from traditional MAS, clarifying analytical granularity and definitional ambiguities. Second, we outline the characteristics of AI-based agents in MAAI, emphasizing the perception–decision–action(–learn) cycle and the role of FMs. Third, we turn to their organization, situating MAAI at the intersection of process automation, agentic IS, and AI. Finally, we synthesize these roots into a layered conceptual framework that extends the MAS tradition into contemporary IS contexts.

From traditional multi-agent systems to multi-agent AI

The interdependence of agent and society design complicates the classification of complex AI-based IS (Kuehl et al., 2022). A single software entity may be internally composed of multiple sub-agents, each meeting autonomy criteria, thereby blurring the distinction between single- and multi-agent systems. Franklin and Graesser (1996) illustrate this with Sumpy, an agent whose layered architecture qualifies it simultaneously as an individual agent and as an MAS. Thus, whether an orchestrated workflow is analyzed as one composite agent or as an MAS depends on the chosen analytical granularity. Within the IS domain, however, such distinctions are less clear-cut, and boundaries between complex AI-based IS and MAS remain contested (Collins et al., 2021; Dorri et al., 2018; Hofmann et al., 2024; Sapkota et al., 2026).

AI-based agents in multi-agent AI

AI-based agents in MAAI are grounded in established theories of intelligent and autonomous agents. Russell et al. (1995) describe an *intelligent agent* as an entity that perceives its environment through sensors (perception) and acts (action) upon it through actuators to achieve goals, highlighting the perception–action loop as a fundamental cycle. Franklin and Graesser (1996) refine this to define an *autonomous agent* as a system embedded in an environment that continuously senses (perception), acts (action), and adapts over time (decision/learn). Wooldridge and Jennings (1995) characterize agents by autonomy (decision), reactivity (perception), proactiveness (decision/action), and social ability (action/coordination), while Ferber and Weiss (1999) frame them as physical or virtual entities that perceive (perception), act (action), communicate (action/coordination), and pursue goals (decision) within evolving environments. Wooldridge (2009, p.15) synthesizes these perspectives as "a computer system that is situated in some environment, and that is capable of autonomous action in this environment in order to meet its design objectives" (perception–decision–action cycle).

Building on this lineage, AI-based agents in MAAI operationalize the classical perception–decision–action(–learn) cycle but are both enhanced and constrained by the cognitive capabilities (Vernon et al., 2007) of AI-enabled IS (Kuehl et al., 2022). In practice, these capabilities are frequently provided through FMs (Bommasani et al., 2021; Schneider et al., 2024), which are often employed "as is" without further fine-tuning for specific tasks—contrasting with traditional reinforcement learning (RL) agents that iteratively adapt their policies to the environment. Crucially, as Hofmann et al. (2024) cautions, autonomy should not be conceived as a binary property. Instead, it spans a spectrum from automated to semi-autonomous to fully autonomous systems, depending on their self-sufficiency, constraints in fulfilling goals, and networking capacities. This spectrum is essential for positioning MAAI: AI-based agents inherit MAS principles while emphasizing the contemporary flexibility, adaptivity, and coordination enabled by FMs (Bommasani et al., 2021; Liu et al., 2025; Schneider et al., 2024).

Organization of AI-based agents

To address definitional uncertainties, we situate MAAI at the intersection of three foundational IS research streams—*process automation*, *agentic IS*, and *AI*—which are deeply interdependent in digital market ecosystems. Their convergence can be meaningfully interpreted through the interaction-, environment-, and organization-oriented views of the vowel paradigm (da Silva & Demazeau, 2002; Demazeau, 1997).

The first foundational stream, **process automation**, contributes to the environment-oriented view by providing the structural substrate: it traditionally involves transferring work that is organized into processes consisting of events, tasks, and decision points from humans to technology (Benbya et al., 2020; Engel et al., 2022), often to enhance productivity and reduce manual workload (Autor et al., 2003; Hofmann et al., 2020). RPA exemplifies this, using predefined business rules to automate routine tasks (Hofmann et al., 2020; IEEE Guide for Terms and Concepts in Intelligent Process Automation, 2017; Jimenez-Ramirez et al., 2019). It is especially effective in repetitive tasks (Haase et al., 2024; Willcocks et al., 2017). Yet, traditional IS theory has tended to frame such technologies as passive artifacts under human control (Baird and Maruping, 2021). This framing is increasingly challenged as process automation is augmented with AI capabilities, expanding its scope from deterministic rule-following toward cognitive automation that integrates probabilistic decision-making, adaptive classification, and context-sensitive responses (Engel et al., 2022; Hofmann et al., 2020; Janiesch et al., 2021). When embedded within MAAI, process automation thus shifts from predictable orchestration in bounded environments toward dynamic and reconfigurable workflows, where agents can flexibly adapt to novel situations, negotiate interdependencies, and coordinate across heterogeneous socio-technical contexts (Liu et al., 2025).

The second foundational stream, **agentic IS**, contributes to the interaction-oriented view by foregrounding delegation, coordination, and negotiation among heterogeneous agents, which are either human or computational (Holldack et al., 2026). Reflecting on agentic IS, Baird and Maruping (2021) developed the IS delegation framework. This framework defines agentic IS through three constructs: autonomous agents, their relational mechanisms (e.g., delegation), and the tasks they collaboratively pursue. At the core of agentic IS are agents, human or computational, which are autonomous entities capable of perceiving and acting within an environment, making independent decisions, and interacting with each other (Russell et al., 1995). In doing so, they cooperate and occasionally conflict to achieve individual or collective goals (Baird and Maruping, 2021). Agentic IS artifacts challenge traditional IS views of passive technologies by autonomously initiating actions, managing tasks, and assuming roles and responsibilities (Baird & Maruping, 2021; Feuerriegel et al., 2024). In this context, Baird and Maruping (2021) propose rethinking IS theory to incorporate AI artifacts as agentic actors. Similarly, Schuetz and Venkatesh (2020) argue that cognitive, agentic systems exceed the explanatory scope of existing models of user-information technology interaction, as they can adapt, perceive their environment, and interact with humans and technologies. While Baird and Maruping (2021) primarily focus on dyads of human and agentic IS artifacts, configurations involving multiple interacting artifacts align with MAS (Dorri et al., 2018). However, definitional ambiguity persists. Some scholars treat agentic systems broadly, encompassing single- and multi-agent setups (Baird and Maruping, 2021), while others reserve "multi-agent" for systems with multiple AI agents operating in collaboration (Hughes et al., 2025). This inconsistency is exacerbated by competing labels: Baird and Maruping (2021) use the term "agentic IS artifacts," while Schuetz and Venkatesh (2020) prefer "cognitive computing systems." Other works refer to "AI agents" or "MAS" (Göldi & Rietsche, 2024; Hughes et al., 2025), or to more specific labels such as "large language model (LLM)-based" (Xi et al., 2025), contributing to conceptual ambiguity, communication challenges, and fragmentation across IS discourse.

For MAAI, agentic IS offers essential design principles: delegation, autonomy, and collaboration. These principles articulate how complex tasks can be distributed among multiple autonomous entities and coordinated through relational mechanisms, aligning naturally with the interaction-oriented dimension of the vowel paradigm. They also foreground the need for governance structures—protocols for coordination, conflict resolution, and oversight—when autonomy is distributed across many agents. While the IS delegation framework (Baird and Maruping, 2021) provides a pivotal foundation by centering on the human-agent dyad and its delegation mechanisms, our MAAI framework extends this perspective to the orchestration of multiple AI-based agents, moving beyond dyadic delegation toward multi-agent coordination in complex socio-technical environments.

The third foundational stream, **AI**, expands both the organization- and environment-oriented views by furnishing agents with cognitive capabilities for perception, reasoning, and adaptive behavior (Berente et al., 2021; Kuehl et al., 2022). In the environment-oriented view, AI enables agents to interpret unstructured inputs, learn from data, and adapt actions to dynamic contexts, thereby extending process automation beyond deterministic routines into probabilistic and context-sensitive domains (cognitive automation) (Engel et al., 2022; Hofmann et al., 2020; Janiesch et al., 2021). In the organization-oriented view, advances in FMs have enlarged the feasible scope of agent roles: agents can not only execute

delegated tasks within workflows but also flexibly reorganize processes, negotiate responsibilities, and coordinate with other agents in evolving structures (Banh & Strobel, 2023; Berente et al., 2021; Feuerriegel et al., 2024). Similarly, agentic IS also benefits from advancements in AI, which augment the autonomy and intelligence of agentic IS artifacts (Feuerriegel et al., 2024; Kuehl et al., 2022). AI technologies equip software entities with cognitive functions, enabling them to perform complex knowledge work traditionally requiring human judgment (Berente et al., 2021). Recent developments in generative AI further extend these capabilities. Generative models demonstrate human-like reasoning, decision-making, and adaptive task execution—qualities essential for complex knowledge work (Banh & Strobel, 2023; Feuerriegel et al., 2024). However, these skills remain confined to predefined process points and lack the interactive, decentralized coordination needed for more complex collaborative work. This points to the phenomenon of MAAI: multiple AI-based agents in MAS, widely regarded as the next frontier in the automation of knowledge work (Guo et al., 2024).

The nature of multi-agent AI

Bringing these streams together, MAAI is best understood as an architectural paradigm rather than a singular technology. It represents a resurgence and synthesis of long-standing ideas from MAS, AI, and IS research concepts. What distinguishes the current moment is the maturity of AI techniques, especially FMs. These advances make it far more feasible to construct ensembles of AI-based agents that perceive, reason, act, (and learn) collaboratively, moving beyond the rigid, rule-based scripts of earlier automation toward dynamic, reconfigurable, and knowledge-intensive workflows.

Our choice of the term "Multi-agent AI" is deliberate. While "agentic AI" has gained traction in industry discourse, it risks overstating novelty and obscuring continuity with MAS research. The phrase also underemphasizes the plurality of agents, which is central to the phenomenon. By contrast, MAAI explicitly anchors these developments in the MAS tradition, acknowledging that many so-called "agentic" innovations are rediscoveries of long-established principles (Sapkota et al., 2026) (cf. Table 1 in the Appendix). Framing the phenomenon as MAAI ensures that new AI-based agent collectives are understood as building upon—not reinventing—the conceptual and technical scaffolding of MAS.

From the convergence of process automation (structure), agentic IS (interaction), and AI (cognition) and drawing explicitly on the agent-, interaction-, environment-, and organization-oriented views of the vowel paradigm (da Silva & Demazeau, 2002; Demazeau, 1997), we derive five interdependent layered components (abbreviated as C1 to C5) that characterize MAAI. Figure 1 visualizes these components in an "onion-like" hierarchy. Each layer represents a debate-space cluster formed at the intersection of IS research streams and vowel paradigm views defining MAS. The *FM layer (C1)* corresponds to the *agent-oriented* view by defining the cognitive and representational capabilities of AI-based agents. The *data-centric perception and action layer (C2)* links to the *environment-oriented* view by specifying how agents sense and act upon digital environments. *Dynamic orchestration (C3)* reflects the *interaction-oriented* view through mechanisms of coordination, delegation, and emergent task structuring among agents. The *agent-integrated workflow layer (C4)* aligns with the *organization-oriented* view by embedding agent activity into institutional processes, governance structures, and routinized work practices. Finally, the *interaction interface (C5)* connects these layers back to human actors, enabling meaningful oversight, collaboration, and accountability. Together, these components extend the MAS tradition of agent and society design into IS contexts, embedding micro-level autonomy within macro-level structures of coordination, governance, and human interaction. The following section builds on this foundation and introduces a conceptual framework that characterizes MAAI in light of its conceptual roots.

Multi-agent AI

To understand MAAI and its role in automating complex knowledge work, it is essential to move beyond traditional process automation with its reliance on rigid, predefined workflows. Whereas such systems execute predetermined scripts, MAAI represents a qualitatively different class of intelligent systems (Hofmann et al., 2024): it coordinates multiple autonomous entities that collaborate dynamically in socio-technical environments.

Definition

Multi-agent artificial intelligence (MAAI) *refers to a system architecture composed of multiple autonomous or semi-autonomous AI-based agents that collaboratively perceive, reason, and act to perform context-sensitive, cognitive tasks. These agents dynamically structure workflows, coordinate roles across heterogeneous entities, including humans, and interact through interfaces within socio-technical environments.*

While our definition draws on the classic perception–decision–action cycle of agents (Russell et al., 1995; Wooldridge, 2009), it is important to note a key difference between AI-based agents of MAAI grounded in FMs (FM-based agents) and classical RL agents. Traditional RL agents iteratively adapt their policies through environmental feedback. In con-

trast, FM-based agents inherit most of their capabilities from large-scale pre-training and can only "learn" in limited ways (e.g., via fine-tuning with direct preference optimization) (Rafailov et al., 2023). Thus, their "learning" is not continuous adaptation in situ, but rather the recombination and contextualization of pre-trained knowledge. This constraint is crucial for IS research, since it highlights both the scalability of FM-based agents and their dependency on external updates or organizationally controlled fine-tuning.

The definition specifies *what* constitutes MAAI, but it does not yet clarify *how* such systems operate in practice. To address this, we conceptualize MAAI as a layered architecture of five interdependent components. These components encapsulate AI capabilities in a nested, "onion-like" structure (Fig. 1), enabling AI-based agents to collectively perceive, reason, and act in ways that extend far beyond the rigid, rule-based logic of RPA. Importantly, MAAI systems do not merely execute fixed workflows; rather, they are designed to dynamically construct agent-integrated workflows that emerge through interaction with data, other agents, and human actors in the environment. The five components of the framework, *foundation model (C1), data-centric perception and action (C2), dynamic orchestration (C3), agent-integrated workflow (C4),* and *interaction interface (C5)*, provide a structured lens to disentangle the complexity of MAAI. By mapping each component to its conceptual roots in process automation, agentic IS, and AI, the framework both explains the mechanisms of MAAI and establishes a theoretically grounded basis for systematic investigation in IS research.

To enhance accessibility and illustrate its practical relevance, we complement the framework with a use case situated in digital markets: an insurance company's handling of customer inquiries regarding policy coverage for vehicle damage (Fig. 2). This example, situated in the domain of digital markets, demonstrates the framework's capability to respond autonomously to customer inquiries regarding policy coverage for vehicle damage. Specifically, the automation involves the framework's components to determine whether the car damage claim falls within the scope of a customer's insurance policy. The customer submits the inquiry through a digital platform, for example, a mobile insurance app, including a textual description and images of the damaged cars and scenery.

Component C1: Foundation model

The base of the framework rests on FMs, which refer to a class of large-scale AI models, pre-trained on vast amounts of data, that serve as a versatile foundation for a variety of downstream tasks (Bommasani et al., 2021; Liu et al., 2025; Schneider et al., 2024; Wu et al., 2025). Typically based on deep neural networks like transformers, FMs capture complex data relationships and can be fine-tuned for specific tasks or used as is across various applications. FMs are often LLMs or large multimodal models (LMMs), such as GPT-x

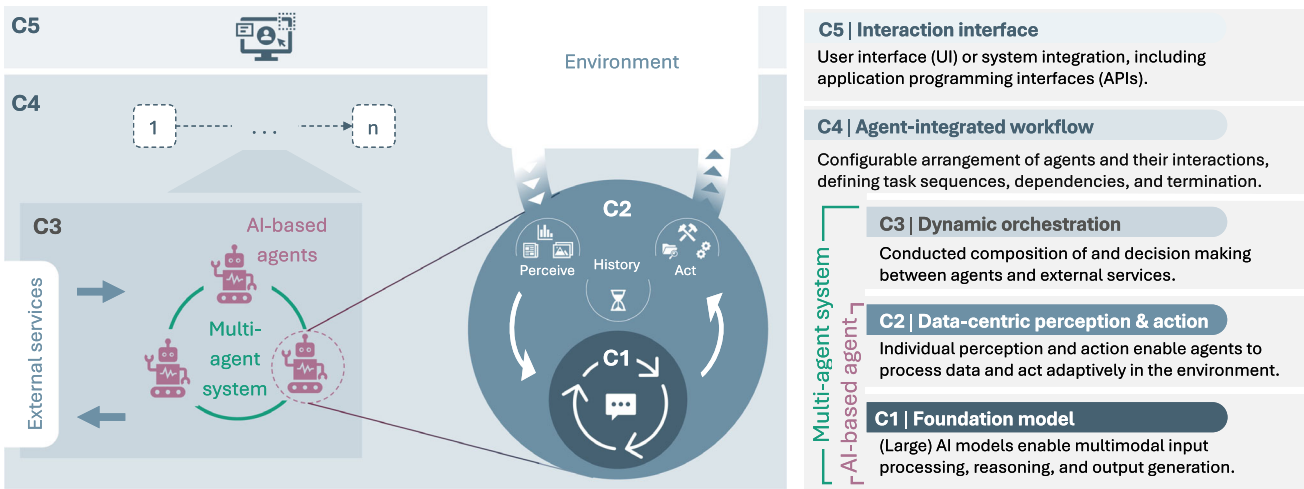

Fig. 1 Multi-agent AI is characterized by a framework composed of a MAS structured along five interrelated components. At its core, *diverse foundation models (C1)*, such as LLMs or LMMs, equip agents with the functionality to process multimodal inputs, integrate background knowledge, perform reasoning, and generate coherent outputs. The *data-centric perception and action (C2)* component enables agents to gather, contextualize, and act upon information from their environment. *Dynamic orchestration (C3)* manages the coordination and decision-making among multiple agents and external services, allowing flexible and adaptive task allocation. The *agent-integrated workflow (C4)* defines the sequence, interdependence, and logic of agent activities. Finally, the *interaction interface (C5)* connects the MAS to its users or external systems—either via user interfaces (UIs) for human-AI collaboration or through application programming interfaces (APIs) for integration with other software systems

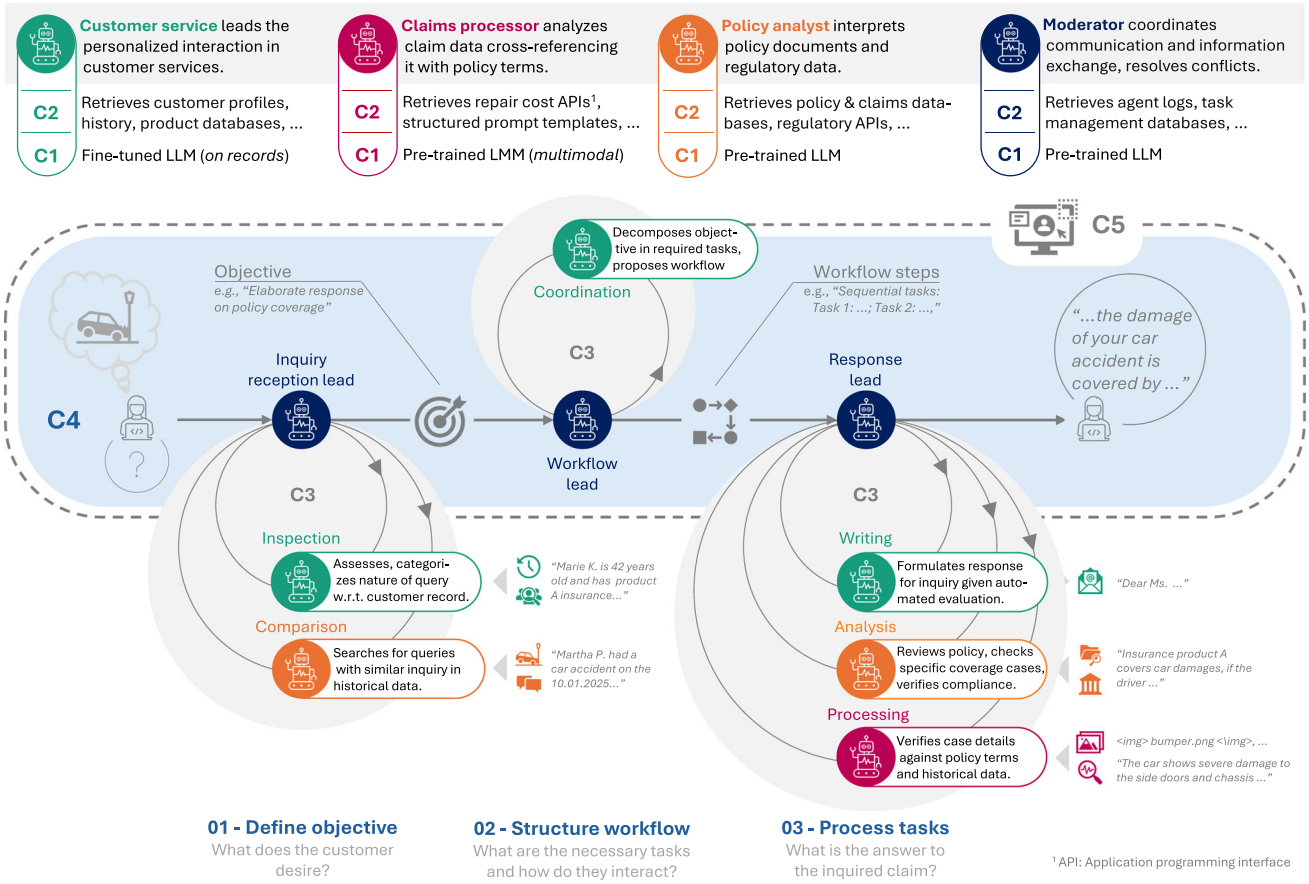

Fig. 2 An exemplary MAAI-based workflow in an insurance company setting. The illustration shows how multiple specialized AI-based agents, each leveraging a *foundation model (C1)* and a *data-centric perception and action component (C2)*, are *dynamically orchestrated (C3)* to form an *agent-integrated workflow (C4)*. The workflow begins by defining the customer's objective—clarifying coverage for car damage—structuring the necessary tasks and agent roles, and subsequently processing those tasks to deliver a comprehensive, policy-compliant response. Throughout this process, the *interaction interface (C5)* facilitates intuitive interaction, ensuring that human stakeholders can monitor or intervene as needed

or Gemini. While LLMs solely provide natural language processing, LMMs also extend the ability to process further data modalities, such as vision or audio data. Both constitute the cognitive capability of AI-based agents in MAAI, equipping them to reason, process unstructured data, and generate coherent outputs (Liu et al., 2025; Lu et al., 2023; Schick et al., 2023). Importantly, Electronic Markets has already published valuable groundwork on machine learning (ML), generative AI, and cognitive automation. For example, Janiesch et al. (2021) introduce ML and deep learning to IS research; Kuehl et al. (2022) propose a framework for intelligent agents; Binder et al. (2022) discuss explainability in LLMs; Banh and Strobel (2023) survey generative AI potentials and challenges; Schmidt et al. (2023) analyze assistant platforms built on FMs; and Hofmann et al. (2024) emphasize the autonomy and transparency issues of FM-based agents. Concretely, three FM properties are particularly salient for IS research and for understanding MAAI (Schneider et al., 2024): first, the emergence of novel capabilities (e.g., in-context learning) (Brown et al., 2020; Min et al., 2022; Webb et al., 2023), which substantially increase agent flexibility while simultaneously introducing forms of unpredictability. Second, FMs drive a homogenization of AI development around a small number of dominant models (Bommasani et al., 2021; Liu et al., 2023; Schneider et al., 2024), lowering entry barriers but also centralizing systemic risks, including the possibility of algorithmic monocultures (Fishman & Hancox, 2022; Kleinberg & Raghavan, 2021). Third, FM-based agents exhibit pronounced prompt sensitivity, where minor variations in input can lead to markedly different outputs, making prompt engineering both a productive affordance and a potential vulnerability (George et al., 2025; Gutheil et al., 2025; Wei et al., 2023).

In our exemplary insurance use case, FMs are crucial for handling the increasing complexity and data volume. Their scalability and generalization capabilities enable them to manage large and diverse inquiries—ranging from simple coverage questions to complex, multimodal inquiries—while

drawing on a broad, pre-trained knowledge base. FMs, particularly LMMs, can seamlessly process and integrate various data types (e.g., textual claims or images of car damage) within a unified system, eliminating the need for different specialized AI models. Additionally, their ability to be fine-tuned allows them to adapt to the insurance particularities, ensuring long-term flexibility and relevance, such as standards and style of customer responses.

Component C2: Data-centric perception and action

Building upon FMs, AI-based agents comprise perception and action, composing C2 to interact with their environment (Kuehl et al., 2022; Xi et al., 2025). In line with prior research in CS, we define agents as entities that are modeled using concepts traditionally associated with human characteristics, such as autonomy, social ability, reactivity, and proactiveness (Goodwin, 1995; Wooldridge & Jennings, 1995). Rooted in MAS and informed by agentic IS concepts, AI-based agents extend traditional process automation by incorporating human-centric capabilities like collaboration, decision-making, and adaptability. Consequently, each AI-based agent must incorporate mechanisms to perceive and act upon its environment, grounded in a data-centric approach (Jakubik et al., 2024). This approach acknowledges that while the underlying FM—its architecture and capacities—remain fixed, its effective adaptation and specialization are based on information in the form of data it receives and processes.

In data-centric AI, the primary idea for improving system performance and tailoring behaviors lies not in modifying the AI models' architecture but rather in refining, selecting, and structuring the data that AI-based agents perceive and use as context (Bhatt et al., 2024). Consequently, the perception component must sense and prepare diverse forms of data, optimizing them for the FM's internal representations. Although the FM's fundamental capabilities remain constant, the agent's skillfulness, domain relevance, and ability to handle evolving tasks are shaped by the data that it ingests and transforms into actionable insights. Since FMs store their learned representations in architectural parameters (weights) obtained from extensive prior training, they only influence their performance by altering the data they process, both as input and through the retrieval of external information. Perception in MAAI operates entirely on data: it transforms raw environmental inputs into structured, coherent formats that align with the FM's learned latent space of representations. This includes external data from sensors, documents, user inputs, application programming interfaces (APIs), and other sources, as well as internally managed context (e.g., previously generated text or prompt-engineered instructions). Through techniques like retrieval-augmented generation (RAG) (Klesel & Wittmann, 2025; Lewis et al., 2020; Li et al., 2025), active feature acquisition (Hübotter et al., 2024), structured templates (Wei et al., 2022; Yang et al., 2025), and APIs (Liang et al., 2024), the perception component curates and enhances the data delivered to the FM, effectively guiding the model's output without changing its core parameters. AI-based agents can access external databases or knowledge repositories through these mechanisms, retrieving relevant information dynamically to augment the context available to the FM. Here, the choice and quality of data—which documents to retrieve, what contexts to include, and how to structure the information—enforce distinct characteristics of the AI-based agents.

Once the FM processes the perceived data and produces an output, the action component interprets this output and executes actions in a data-mediated environment. Actions themselves are represented and performed through data: producing a textual response, generating a visual artifact, calling an API, or interacting with external systems are all rendered as data flows. This concept often involves leveraging specialized tools that enable the FM to perform specific functions, such as engaging in vocal conversations. By conceptualizing perception and action as data-centric processes—where even the agent's "touch" on its environment occurs through data-level transformations and interfaces—agents operate within and manipulate data representations. In addition to perception and action, C2 contains the capability to store and manage data, thereby increasing the contextual richness available to the FM. This stored context provides continuity across tasks, enabling AI-based agents to refine their decisions and responses based on historical and situational data.

Consider again the workflow of an insurance company handling a customer's car damage claim (Fig. 2). Each specialized AI-based agent—be it the policy analyst, claims processor, or customer service agent—derives its expertise and context from both static and dynamic sources. Static elements such as predefined roles, agent profiles, and base prompts encode task-specific behavior and domain knowledge. Dynamically retrieved data, such as policy documents, customer records, or regulatory APIs, are then integrated during task execution to contextualize perception and guide action. For instance, the policy analyst can interpret insurance clauses by retrieving relevant documents, while the customer service agent adjusts tone and content by drawing on conversation history, customer records, and domain-specific prompts. After processing this curated input data, the FM's output directs the subsequent actions, enabling context-specific responses such as generating reports, sending messages, or interacting with external tools. These actions, too, remain data-driven: both perception and execution are shaped by what data can be accessed, structured, and operationalized in the agent's environment.

Component C3: Dynamic orchestration

To effectively integrate the outputs and actions of various agents and maintain coherence within the agent-integrated workflow, AI-based agents must possess a distinct identity that grants them social standing (Bisk et al., 2020). This identity is essential for enabling social cooperation and competition within an MAS. The implementation of such a system comprises a registry of role-specific agents, each designed to be selected and deployed based on the demands of the task at hand (Li et al., 2025; Wang et al., 2024). This composition must be dynamically orchestrated in alignment with the specific requirements of the workflow. The orchestration should be sufficiently adaptable, allowing each workflow to autonomously determine its optimal configuration of agents. This ensures the orchestration can address the full complexity inherent in knowledge-intensive tasks. The orchestrated agents' capabilities are extended by connecting external services such as messaging applications or payment tools. In line with previous architectures like service-oriented computing, this integration allows orchestrated agents to execute actions that require interaction with external systems, broadening the scope of tasks they can handle (Calisti et al., 2010; Liu et al., 2025; Tapia et al., 2010).

In the context of MAS, the division of labor principle underpins the specialization of agents, where each agent, equipped with domain-specific expertise, is optimized for executing particular tasks. This specialization enhances the efficiency and precision of task execution, as it minimizes the overhead associated with task switching by decomposing complex workflows into discrete, manageable subtasks. Xi et al. (2025) delineate agent interactions into two primary paradigms: cooperative interaction for complementarity and adversarial interaction for strategic advancement. Cooperative interaction involves agents that continuously evaluate their counterparts' operational states and capabilities, engaging in collaborative task execution and knowledge exchange (Li et al., 2023). This interaction can be either disordered or ordered. In disordered cooperation, multiple agents operate with a high degree of autonomy, freely contributing their inputs and feedback within the system, which may result in misaligned objectives and potential disruptions in workflow orchestration (Mandi et al., 2024; Xi et al., 2025). Conversely, ordered cooperation adheres to predefined communication protocols, where agents sequentially process information, with downstream agents dependent on the outputs of upstream counterparts (Li et al., 2023, 2024). Dynamic orchestration mechanisms, such as a moderator agent, can be deployed to manage such interactions effectively to regulate communication flows and ensure adherence to structured review processes. In adversarial settings, agents dynamically reconfigure their strategies in response to real-time environmental feedback and the actions of competing agents (Abdelnabi et al., 2024). This adaptive behavior is crucial in adversarial MAS, where the objective is to optimize decision-making and enhance the quality of responses in high-stakes scenarios (Xi et al., 2025). AI-based agents may independently pursue optimization objectives, such as utility maximization, which aligns with principles of game theory (Mao et al., 2023). However, to maintain a structured competitive environment, deploying a moderator agent can facilitate the orchestration of interactions, ensuring that the competitive dynamics remain aligned with the overall system objectives.

In our insurance company use case, dynamic orchestration is essential for managing cooperative agent interactions. Cooperative interactions allow agents to independently contribute towards aligning on a common objective, while enabling the system to flexibly adapt the workflow to the situation at hand. In a simple coverage inquiry, the moderator agent orchestrates a minimal workflow: the policy analyst retrieves and interprets the relevant policy clauses, the claims processor cross-checks the submitted information against historical cases, and the customer service agent drafts the final explanation. Once the required information is aligned, the moderator terminates the workflow. By contrast, a higher-complexity case with potential fraud indicators triggers a markedly different, dynamically assembled workflow. Here, the customer service agent analyzes anomalies in the submitted images, inconsistencies in timestamps, or mismatches with prior claims. This agent may engage in multiple iterative loops: requesting further evidence from specialized fraud detection agents, re-evaluating policy constraints, or escalating the case to a human fraud investigator. Throughout this process, the moderator adapts the orchestration strategy in real time, deciding which agents to involve, how to sequence interactions, and when to switch from cooperative to more adversarial evaluation patterns. Across both scenarios, the moderator agent regulates the information flow, resolves conflicts between partial outputs, and aligns the contributions of all involved agents.

Component C4: Agent-integrated workflow

Given the inherent flexibility and complexity of many real-world tasks, they necessitate an agent-integrated workflow and dynamic decision-making process, which may be automatically created within MAAI (Liu et al., 2025). Rooted in the "information pull" paradigm, this component is conceptualized as a dynamic workflow, generated in direct response to emergent information needs (Cichocki et al., 1998). In the context of MAAI, this dynamic workflow is operationalized through the orchestration of AI-based agents, which execute tasks guided by abstract design patterns. Van der Aalst et al. (2003) introduced a spectrum of design patterns, ranging from elementary constructs to intricate abstractions. The sim-

pler design patterns are generally aligned with sequential or parallel workflows, facilitating the pre-definition of workflow steps and enabling task-oriented deployment through high-level directives (Xi et al., 2025). This approach facilitates the orchestration to establish all requisite tasks in advance, allowing AI-based agents to execute them autonomously. Each task's outcome is then sequentially forwarded to the subsequent task. Conversely, more intricate workflows exploit the full intellectual potential of MAS through sophisticated orchestration. These workflows incorporate iterative loops, necessitating innovation-oriented deployment or even lifecycle-oriented deployment, enabling agents to continuously interact with an open and unknown environment (Xi et al., 2025).

In our example, the customer's inquiry is addressed through a three-stage process: initially, the overall objective is identified, followed by the structuring of the workflow, and culminating in the processing of the tasks, all within the collaborative framework facilitated by the dynamic orchestration of the MAS. Each stage supports dynamic orchestration and decision-making, allowing for adaptive and responsive interactions throughout the workflow. A policy analyst agent might need to query databases an unknown number of times to gather sufficient information about a case. In such a complex scenario, the agent-integrated workflow component showcases innovation-oriented deployment, iteratively adjusting tasks as the process evolves. This method enhances flexibility and adaptability within workflow automation, accommodating changes and enabling dynamic decision-making processes that emulate human-like deliberations rather than adhering to rigid rule-based protocols.

Component C5: Interaction interface

The interaction interface serves as the entry point for both human and system-level engagement with MAAI. It encompasses not only user interfaces (UIs) but also APIs, enabling integration with external systems and organizational platforms. While the previous components enable AI-based agent coordination and agent-integrated workflow generation, the interaction interface ensures that these processes are accessible, traceable, and actionable. As such, the UI constitutes the critical socio-technical junction where the system becomes actionable in organizational settings. Effective UIs are also essential for enabling human oversight and intervention, especially in complex, automated workflows (Sterz et al., 2024). Drawing on IS perspectives (Baird & Maruping, 2021; Schuetz & Venkatesh, 2020), this component plays a dual role: supporting human-AI collaboration through explainable, adaptive, and role-specific interfaces, and facilitating machine-to-machine communication through structured endpoints. In organizational settings, such interfaces must balance transparency with usability—offering dashboards, conversational agents, or monitoring tools for internal users, and streamlined interaction channels for external systems. This goes beyond the necessity of having any interaction interface—it raises design requirements related to explainability, adaptive feedback, and trust-building mechanisms in human-AI collaboration. Designing an appropriate interaction interface for MAAI thus depends on the system's purpose and user roles. For end-users, chatbot interfaces or decision dashboards may suffice to deliver responses and gather input. For internal users (e.g., employees monitoring agent-integrated workflows), the UI must support traceability of agent interactions, intervention capabilities, and contextual awareness. In both cases, prior work in IS suggests the importance of adaptive, stateful interfaces and role-specific views to manage complexity and promote trust (Gimpel et al., 2020; Gnewuch et al., 2017).

In the insurance company use case, the interaction interface connects customers, employees, and the MAAI. A chatbot UI allows customers to submit claims and receive updates, while internal APIs dynamically feed outputs from policy analysts or claims processors into case management systems. These interaction interfaces ensure that the system remains comprehensible, responsive, and compatible with existing workflows.

Challenges and emerging pathways for the IS community in electronic markets

The advent of modular agent-based AI systems—operating within organizations and across digital platforms, multi-sided marketplaces and networked business environments—marks a shift from conventional, narrowly defined automation toward platform-embedded agentic architectures that contribute to value creation, exchange mechanisms and business-ecosystem governance. Early research in electronic markets explored agent-based coordination, for example, autonomous buyers, sellers, and intermediaries negotiating prices (Kehagias et al., 2005; Oliver, 1996) or recommending products (Benbasat & Wang, 2005; Xiao & Benbasat, 2007). However, only recently have FM-enabled MAS entered this domain, bringing qualitatively new capabilities. In particular, FMs (Schneider et al., 2024), especially those employed in generative AI (Feuerriegel et al., 2024), represent transformative potential not only for automating knowledge work but for reconfiguring how digital marketplaces and service platforms mediate exchanges among actors.

MAAI can ingest and reason over unstructured and non-standardized data and contextualize it through multiple distributed agents offering different perspectives. This substantially expands what agent-based electronic markets can achieve, especially given the interdependencies and organizational boundaries that characterize modern platforms

and supply networks. In e-commerce and digital platforms, MAAI enables autonomous agents to negotiate, recommend, or coordinate logistics using not only structured catalog data but also customer reviews, images, sensor feeds, or social-media content. Across supply chains, interacting agents can support adaptive coordination optimizing procurement, forecasting demand, or scheduling shipments by integrating heterogeneous data sources (Nambisan, 2000; Singh et al., 2005; Wu et al., 2000). Therefore, MAAI represents a next stage in electronic market design: coordination and value exchange increasingly occur among heterogeneous, semi-autonomous agents capable of rich contextual reasoning, rather than solely among human actors operating through rigid, structured data flows.

For the IS community, and specifically within the domain of networked business and electronic markets, this development demands theoretical reflection and actionable guidance on how MAAI interplays with platform governance, marketplace coordination, business model innovation, and socio-economic integration. The conceptual framework presented in this work seeks to provide such guidance in the context of digital platforms and electronic markets. Each component addresses a core architectural layer of MAAI systems in platform settings, offering a scaffold for identifying requirements, selecting technologies, defining agent roles, and assessing socio-technical implications in marketplace or platform ecosystems. Researchers may use the framework to compare system architectures across platform settings, identify gaps in existing multi-agent solutions within electronic markets, and formulate research questions grounded in component-specific challenges (cf. Fig. 3). Practitioners—especially platform owners, digital marketplace operators, and ecosystem designers—can use it as a reference for designing modular, agent-based systems that enable value exchange, orchestration, and marketplace governance in real-world digital ecosystems. Some research questions naturally link to adjacent fields: those on C1–C2 overlap with work on LLMs, RAG and AI-based services in platforms; those on C4–C5 connect to issues of trust, transparency, human-agent interaction, and governance in electronic markets. By contrast, challenges around C3 (dynamic orchestration) are particularly distinctive for agent ecosystems within platformed or networked business contexts.

Engineer technical capabilities (Components C1–C3)

In the context of digital platforms and networked marketplaces, MAAI's potential to support participants (e.g.,

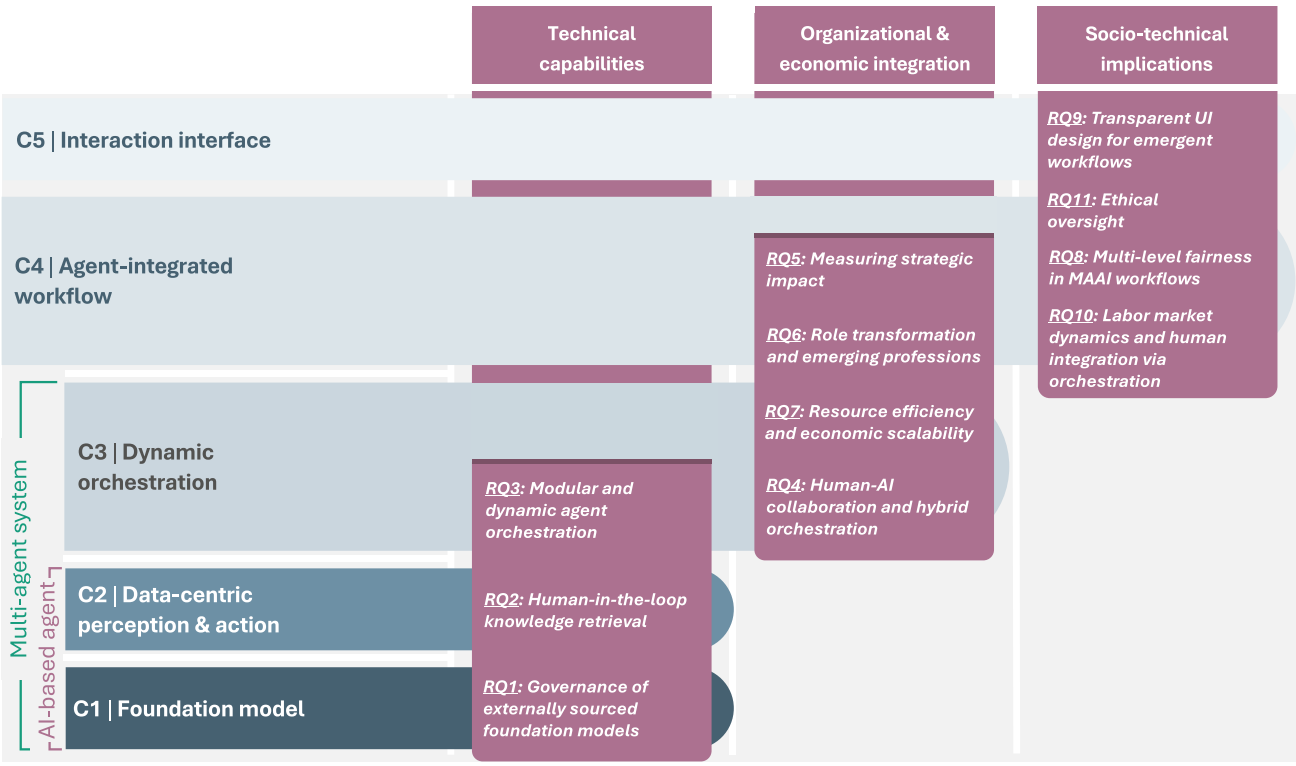

Fig. 3 The figure maps the five-component MAAI framework—*comprising foundation model (C1), data-centric perception and action (C2), dynamic orchestration (C3), agent-integrated workflow (C4),* and *interaction interface (C5)*—to exemplary research questions from the three proposed analytical pathways: technical capabilities, organizational and economic integration, and socio-technical implications

platform stakeholders, marketplace operators, ecosystem partners) and to automate complex exchange and coordination tasks introduces substantial engineering challenges. In particular, the design of coherent agent systems that enable dynamic orchestration (C3) of agents across platform workflows is critical. A central issue is that FMs (C1)—often externally developed and accessed via APIs—limit platform operators' ability to govern the evolution of models, fine-tune them for specific domains (e.g., multi-sided platforms, digital matching services), and mitigate propagated bias in marketplace decisions (e.g., pricing, matching, recommendation) (Fetzer et al., 2025; Ji et al., 2023; Ray, 2023). Within MAAI ecosystems on platforms, a misalignment or faulty output from one agent may propagate through other agents, influencing matchmaking, pricing, service orchestration, or network interactions; thus amplifying risks in organizational workflows and digital market settings. This underscores the need for processes that align model capabilities with platform governance, business model requirements, and value creation outcomes. To address these engineering challenges in networked business contexts, design science research approaches (e.g., Gregor and Hevner, 2013; Peffers et al., 2007; Tuunanen et al., 2024) provide a structured foundation for developing design knowledge around agent-based systems.

However, existing frameworks remain largely conceptual (Händler, 2023; Zhou et al., 2024) and insufficiently prescribe how to implement MAAI in value-generating platform contexts (e.g., multi-sided marketplaces, digital intermediaries) (Chen et al., 2025; Durante et al., 2025). Researchers should therefore develop reference architectures, methodological guidelines, decision models, and modular software components tailored for orchestrating heterogeneous agents (C3) within platform ecosystems, informed by domain-specific constraints of digital marketplaces (e.g., matching logic, pricing algorithms, network effects, complementor coordination). Beyond agent orchestration, the effectiveness of data-centric perception and action (C2) in platform contexts hinges on how agents access, interpret, and act upon relevant data (e.g., transaction logs, user-platform interactions, network flows). Traditional retrieval techniques tend to rely on static documentation, limiting responsiveness in dynamic marketplace environments. There is a need to integrate human-generated insights (e.g., partner behavior, community feedback, platform logs) into retrieval mechanisms (e.g., active feature acquisition) to enhance contextual sensitivity and improve agent performance in dynamic platform ecosystems (Hübotter et al., 2024). These challenges give rise to the following research questions for platform-oriented agent systems:

RQ1 **Governance of externally sourced foundation models (C1)**: Given that FMs are often externally developed and integrated via APIs, leaving development teams with limited control over their evolution, how can organizations establish continuous development and deployment processes that align model capabilities with domain-specific requirements and mitigate risks related to opacity, bias, and functional drift?

RQ2 **Human-in-the-loop knowledge retrieval (C2)**: How can human-generated insights and tacit expertise be systematically integrated into knowledge retrieval mechanisms to enhance contextual relevance and move beyond static sources such as documentation or wikis?

RQ3 **Modular and dynamic agent orchestration (C3)**: How can design principles be applied to enable modular and dynamic orchestration of heterogeneous AI-based agents within MAAI systems, allowing workflows to be adaptively structured and refined in real-time as tasks evolve?

Structure organizational and economic integration (Components C3–C4)

From a managerial and platform-ecosystem standpoint, integrating MAAI into digital marketplace workflows invites fundamental questions about the future of platform-mediated exchange, the evolution of human roles in digital networks, and the changing architecture of coordination across platform stakeholders. AI-based agents are increasingly able to execute complex, multi-step processes (e.g., matching supply and demand, orchestrating complementors, enabling service bundles), yet human involvement remains essential for oversight, governance, exception handling, and network orchestration (Liu et al., 2025). From the perspective of electronic markets, such modular and dynamically orchestrated agent architectures reshape how digital platforms coordinate transactions, mediate markets, structure networked business relationships, and enable value creation across ecosystem participants. Embedding agentic architectures into organizational workflows and platform infrastructures means embedding agency, negotiation, and contextual reasoning into the core of marketplace logic. Dynamic orchestration (C3) thus requires mechanisms that dynamically allocate tasks between human and AI-based agents depending on context, network conditions, value contribution, and ecosystem position (Mu et al., 2024; Song et al., 2025). In multi-sided platform settings, this may involve dynamic matching of human experts, automated agents, and complementary services within the marketplace ecosystem (Hemmer et al., 2025; Pan et al., 2025). The intersection of MAAI and process mining offers a promising avenue in platform contexts: process mining can reconstruct real-time workflows from event logs (e.g., platform transaction logs, agent interaction logs, ecosystem partner activities), making visible how AI-based agents

form, execute, and adapt marketplace workflows in practice (Bemthuis and Lazarova-Molnar, 2025). This supports both research and managerial oversight of emergent orchestration patterns. From an organizational perspective, deploying MAAI within platforms is expected to create new forms of work (e.g., platform-orchestration roles, agent supervision roles), transform existing roles (e.g., marketplace operations, service provisioning, complementor coordination), and alter how value is created and captured across ecosystem participants. Strategic implementation thus requires not only technical integration but also platform governance, network orchestration, workforce strategy, ecosystem alignment, and business model adaptation. Finally, the economic viability of MAAI in platform contexts is closely tied to resource efficiency, cost-benefit of orchestration mechanisms, scalability of agent ecosystems, network effects, and value capture across platform participants. Not all agents require large FMs; platform operators may assign tasks intelligently (C3) based on task complexity, data sensitivity, network position, or complementor strategy, leveraging lightweight models, model parallelisation, quantisation, or knowledge distillation to enhance scalability and cost-effectiveness of agent ecosystems (Zhu et al., 2024). In light of these developments for digital platforms, we identify the following research questions oriented towards networked business:

RQ4 **Human-AI collaboration and hybrid orchestration (C3):** How can humans be integrated as active agents within MAAI, and how can orchestration mechanisms dynamically allocate human and AI-based agents to tasks based on the comparative value of human skills?

RQ5 **Measuring strategic impact (C4):** What metrics and evaluation frameworks can be developed to assess the strategic and financial benefits of integrating orchestrated AI-based agents into real-world organizational workflows? And how do we measure the performance of an MAAI in a business process—e.g., by aggregate outcome, by human satisfaction, or by emergent properties?

RQ6 **Role transformation and emerging professions (C3, C4):** Which tasks and workflows are most impacted by the deployment of MAAI, and what new professional roles (e.g., operations, orchestration design) are likely to emerge as organizations adapt to MAAI?

RQ7 **Resource efficiency and economic scalability (C3):** How can orchestration strategies be optimized to ensure resource-efficient MAAI deployments, balancing task complexity with the use of lightweight models, parallelization, and modular architectures?

Study socio-technical implications (Components C4–C5)

The socio-technical implications of MAAI extend deeply into digital platforms: as AI-based agents assume increasingly complex decision-making roles in workflows and platform ecosystems, these developments bear on labor market transformations within the platform economy, platform governance, and value distribution across ecosystem actors, sustainability of digital networks, trust, and accountability in marketplace interactions. Within platform contexts, automation may democratize access to expert capabilities, but also risk reinforcing power asymmetries between platform owners, complementors, and users, applying downward pressure on wages, eroding returns for generic skill sets, or altering how value is shared in multi-sided markets (Acemoglu & Restrepo, 2020; Agrawal et al., 2023; Dwivedi et al., 2023). From the perspective of networked business, preserving human responsibility and accountability becomes increasingly challenging when agent interactions are embedded in workflows and transactions. Faulty agent outputs may propagate across workflows, affecting, for example, matching outcomes, pricing, trust among actors, and overall ecosystem performance. Transparent evaluation and governance structures are essential for maintaining accountability, fairness, and trust in networked business settings (German Ethics Council, 2023; Ray, 2023; Schemmer et al., 2023). The design of interaction interfaces (C5) becomes critical in marketplace ecosystems: in agent-integrated workflows (C4) where tasks and agent configurations evolve on-the-fly, platform interfaces must provide transparency to human stakeholders (e.g., workers, buyers, sellers, complementors, platform operators) without inducing cognitive overload. Long agent dialogues or agent networks may obscure rather than clarify decision logic, which threatens appropriate reliance and user trust in marketplace settings (Ray, 2023; Schemmer et al., 2023). Research questions for this third pathway (with a focus on digital markets and platform ecosystems) include:

RQ8 **Multi-level fairness in MAAI workflows (C4)**: To what extent should principles of algorithmic fairness be applied uniformly—particularly when certain outputs may not directly interface with human stakeholders?

RQ9 **Transparent UI design for emergent workflows (C5)**: How can interaction interfaces be designed to transparently present the evolving outputs and decisions of MAAI systems, especially in workflows where tasks and agent roles are dynamically generated, and how can such interfaces promote appropriate reliance without overwhelming users with excessive or unintelligible information?

RQ10 **Labor market dynamics and human integration via orchestration (C3, C4)**: How will the widespread adoption of MAAI reshape labor markets, particularly as human roles become dynamically integrated into orchestrated workflows, and what policy interventions or training programs are needed to address resulting changes in employment levels, wage structures, and skill demands?

RQ11 **Ethical oversight (C4, C5)**: How can agent-integrated workflows be designed to maintain transparency, draw boundaries of human responsibility, and prevent the undermining of accountability?

Conclusion and research directions

MAAI builds on decades of research in MAS, process automation, and agentic IS, while extending these traditions into a new technological and organizational context shaped by FM and data-centric AI. By conceptualizing MAAI as a five-layer architecture—*foundation model (C1), data-centric perception and action (C2), dynamic orchestration (C3), agent-integrated workflow (C4),* and *interaction interface (C5)*—we have articulated how cognitive capabilities, coordination mechanisms, and human-AI interfaces interlock in ensembles of AI-based agents. This layered view situates MAAI as more than a rebranding of established concepts: it clarifies how recent advances enable flexible orchestration of knowledge work that is adaptive, socio-technical, and organizationally embedded. Our framework delineates three interdependent pathways for future research. On the technical level, MAAI requires advances in agent modularity, orchestration logics, and validation methods that ensure reliability and robustness when workflows are dynamically composed from opaque FMs. On the organizational level, MAAI opens questions about hybrid human-AI work arrangements, shifting occupational roles, value distribution, and how methods such as process mining can make emergent agentic workflows transparent and optimizable. On the socio-technical level, MAAI raises fundamental concerns of accountability, fairness, and governance, demanding mechanisms that sustain trust and responsible oversight when autonomy and agency are distributed across multiple actors.

By integrating these perspectives, the IS community can shape MAAI not simply as a technical evolution, but as a paradigmatic development in the design of digital systems—one that redistributes intelligence and coordination across human and computational agents. Grounded in its conceptual roots yet attuned to emerging challenges, this perspective positions MAAI as a foundation for cumulative, interdisciplinary research into the future of knowledge work, digital markets, and socio-technical systems.

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References

Abdelnabi, S., Gomaa, A., Sivaprasad, S., Schönherr, L., & Fritz, M. (2024). Cooperation, competition, and maliciousness: LLM-stakeholders interactive negotiation. *Advances in Neural Information Processing Systems* (Vol. 37, pp. 83548–83599). Vancouver, Canada: Curran Associates, Inc.

Acemoglu, D., & Restrepo, P. (2020). Robots and jobs: Evidence from US labor markets. *Journal of Political Economy, 128*(6), 2188–2244. https://doi.org/10.1086/705716

Acharya, D. B., Kuppan, K., & Divya, B. (2025). Agentic AI: Autonomous intelligence for complex goals - A comprehensive survey. *IEEE Access, 13*, 18912–18936. https://doi.org/10.1109/ACCESS.2025.3532853

Agrawal, A., Gans, J. S., & Goldfarb, A. (2023). Do we want less automation? *Science, 381*(6654), 155–158. https://doi.org/10.1126/science.adh9429

Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics, 118*(4), 1279–1333. https://doi.org/10.1162/003355303322552801

Baird, A., & Maruping, L.M. (2021). The next generation of research on IS use: A theoretical framework of delegation to and from agentic is artifacts. MIS Quarterly, *45*(1), 315–341. https://doi.org/10.25300/MISQ/2021/15882

Banh, L., & Strobel, G. (2023). Generative artificial intelligence. *Electronic Markets, 33*, 63, 1–17. https://doi.org/10.1007/s12525-023-00680-1

Bemthuis, R. H., & Lazarova-Molnar, S. (2025). Towards integrating process mining with agent-based modeling and simulation: State of the art and outlook. *Expert Systems with Applications, 281*, 127571. https://doi.org/10.1016/j.eswa.2025.127571

Benbasat, I., & Wang, W. (2005). Trust in and adoption of online recommendation agents. *Journal of the Association for Information Systems*, *6*(3), 4. https://doi.org/10.17705/1jais.00065

Benbya, H., Davenport, T.H., Pachidi, S. (2020). Artificial intelligence in organizations: Current state and future opportunities. *MIS Quarterly Executive*, *19*(4). https://doi.org/10.2139/ssrn.3741983

Benke, I., Knierim, M., Adam, M., Beigl, M., Dorner, V., Ebner-Priemer, U., & Weinhardt, C. (2024). Hybrid adaptive systems. *Business &*

Information Systems Engineering, 66(2), 233–247. https://doi.org/10.1007/s12599-024-00861-y

Berente, N., Gu, B., Recker, J., & Santhanam, R. (2021). Managing artificial intelligence. *MIS Quarterly*, *45*(3), 1433–1450. https://doi.org/10.25300/MISQ/2021/16274

Bhatt, N., Bhatt, N., Prajapati, P., Sorathiya, V., Alshathri, S., & El-Shafai, W. (2024). A data-centric approach to improve performance of deep learning models. *Scientific Reports, 14*(1), 22329. https://doi.org/10.1038/s41598-024-73643-x

Binder, M., Heinrich, B., Hopf, M., & Schiller, A. (2022). Global reconstruction of language models with linguistic rules-explainable AI for online consumer reviews. *Electronic Markets, 32*(4), 2123–2138. https://doi.org/10.1007/s12525-022-00612-5

Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., Lapata, M., Lazaridou, A., May, J., Nisnevich, A., Pinto, N., & Turian, J. (2020). Experience grounds language. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 8718–8735). Association for Computational Linguistics.

Bock, A. C., & Frank, U. (2021). Low-code platform. *Business & Information Systems Engineering, 63*(6), 733–740. https://doi.org/10.1007/s12599-021-00726-8

Bommasani, R., Hudson, D.A., Adeli, E., Altman, R., Arora, S., Arx, S.v., Bernstein, M.S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., Buch, S., Card, D., Castellon, R., Chatterji, N., Chen, A., Creel, K., Davis, J.Q., Demszky, D., Donahue, C., Doumbouya, M., Durmus, E., Ermon, S., Etchemendy, J., Ethayarajh, K., Fei-Fei, L., Finn, C., Gale, T., Gillespie, L., Goel, K., Goodman, N., Grossman, S., Guha, N., Hashimoto, T., Henderson, P., Hewitt, J., Ho, D.E., Hong, J., Hsu, K., Huang, J., Icard, T., Jain, S., Jurafsky, D., Kalluri, P., Karamcheti, S., Keeling, G., Khani, F., Khattab, O., Koh, P.W., Krass, M., Krishna, R., Kuditipudi, R., Kumar, A., Ladhak, F., Lee, M., Lee, T., Leskovec, J., Levent, I., Li, X.L., Li, X., Ma, T., Malik, A., Manning, C.D., Mirchandani, S., Mitchell, E., Munyikwa, Z., Nair, S., Narayan, A., Narayanan, D., Newman, B., Nie, A., Niebles, J.C., Nilforoshan, H., Nyarko, J., Ogut, G., and Orr, L., Papadimitriou, I., Park, J.S., Piech, C., Portelance, E., Potts, C., Raghunathan, A., Reich, R., Ren, H., Rong, F., Roohani, Y., Ruiz, C., Ryan, J., Ré, C., Sadigh, D., Sagawa, S., Santhanam, K., Shih, A., Srinivasan, K., Tamkin, A., Taori, R., Thomas, A.W., Tramèr, F., Wang, R.E., Wang, W., Wu, B., Wu, J., Wu, Y., Xie, S.M., Yasunaga, M., You, J., Zaharia, M., Zhang, M., Zhang, T., Zhang, X., Zhang, Y., Zheng, L., Zhou, K., & Liang, P. (2021). *On the opportunities and risks of foundation models*. arXiv. https://doi.org/10.48550/arXiv.2108.07258

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., & Amodei, D. (2020). Language models are few-shot learners. *Advances in Neural Information Processing Systems* (Vol. 33, pp. 1877–1901). Curran Associates, Inc.

Calisti, M., Leymann, F., Dignum, F.P., Kowalczyk, R., & Unland, R. (2010). Service oriented architecture and (multi-)agent systems technology. *Dagstuhl Seminar Proceedings* (Vol. 10021, pp. 1–16). Wadern, Germany.

Chen, W., You, Z., Li, R., Guan, Y., Qian, C., Zhao, C., Yang, C., Xie, R., Liu, Z., & Sun, M. (2025). *Internet of agents: Weaving a web of heterogeneous agents for collaborative intelligence*. "*Proceedings of the Thirteenth International Conference on Learning Representations (ICLR)*". Retrieved from https://openreview.net/forum?id=o1Et3MogPw

Cichocki, A., Helal, A., Rusinkiewicz, M., & Woelk, D. (1998). *Workflow and process automation: Concepts and technology*. London, UK: Kluwer Academic Publishers.

Collins, C., Dennehy, D., Conboy, K., & Mikalef, P. (2021). Artificial intelligence in information systems research: A systematic literature review and research agenda. *International Journal of Information Management, 60*, 102383. https://doi.org/10.1016/j.ijinfomgt.2021.102383

da Silva, J.L.T., & Demazeau, Y. (2002). Vowels co-ordination model. *International Conference on Autonomous Agents and Multiagent Systems* (p. 1129–1136). New York, NY, USA: Association for Computing Machinery.

Davenport, T.H., Jarvenpaa, S., & Beers, M. (1996). Improving knowledge work processes. *Sloan Management Review*, 37, 53–66, Retrieved from https://sloanreview.mit.edu/article/improving-knowledge-work-processes/

Davenport, T. H., & Prusak, L. (1998). *Working knowledge: How organizations manage what they know* (Vol. 5). Boston, MA, USA: Harvard Business School Press.

Demazeau, Y. (1997). Steps towards multi-agent oriented programming. *First International Workshop on Multi Agent Systems (IWMAS'97)*. Boston, MA, USA.

Dorri, A., Kanhere, S. S., & Jurdak, R. (2018). Multi-agent systems: A survey. *IEEE. Access, 6*, 28573–28593. https://doi.org/10.1109/ACCESS.2018.2831228

Drucker, P. F. (1999). Knowledge-worker productivity: The biggest challenge. *California Management Review, 41*(2), 79–94. https://doi.org/10.2307/41165987

Durante, Z., Gong, R., Sarkar, B., Wake, N., Taori, R., Tang, P., Durante, Z., Sarkar, B., Gong, R., Taori, R., Noda, Y., Tang, P., Adeli, E., Lakshmikanth, S. K., Schulman, K., Milstein, A., Terzopoulos, D., Famoti, A., Kuno, N., Llorens, A., Vo, H., Ikeuchi, K., Fei-Fei, L., Gao, J., Wake, N., Huang, Q. & Gao, J. (2025). An interactive agent foundation model. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (p. 3691–3701). Nashville, Tennessee, USA.

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., & Kar, A.K.,.Wright, R. (2023). Opinion paper: "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*, 102642. https://doi.org/10.1016/j.ijinfomgt.2023.102642

Engel, C., Ebel, P., & Leimeister, J. M. (2022). Cognitive automation. *Electronic Markets, 32*(1), 339–350. https://doi.org/10.1007/s12525-021-00519-7

Ferber, J., & Weiss, G. (1999). *Multi-agent systems: An introduction to distributed artificial intelligence* (Vol. 1). Addison-Wesley Reading.

Fetzer, D., Gimpel, H., Meindl, O., & Strickmann, J. (2025). Responsible engineering of information systems based on generative artificial intelligence: An action design research study at a german premium car manufacturer. *Business & Information Systems Engineering*. https://doi.org/10.1007/s12599-025-00950-6

Feuerriegel, S., Hartmann, J., Janiesch, C., & Zschech, P. (2024). Generative AI. *Business & Information Systems Engineering, 66*(1), 111–126. https://doi.org/10.1007/s12599-023-00834-7

Fishman, N., & Hancox-Li, L. (2022). Should attention be all we need? The epistemic and ethical implications of unification in machine learning. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency* (pp. 1516–1527). Seoul, South Korea: Association for Computing Machinery.

Franklin, S., & Graesser, A. (1996). Is it an agent, or just a program?: A taxonomy for autonomous agents. *International Workshop on Agent Theories, Architectures, and Languages* (pp. 21–35). Berlin, Heidelberg: Springer Berlin Heidelberg.

George, J.J., Kotikela, S., & George, T. (2025). Do LLMs care how you ask? Prompt tones and AI accuracy, trust, and engagement. *Proceedings of the International Conference on Information Systems*

(ICIS). Nashville, Tennessee, USA: Association for Information Systems.

German Ethics Council. (2023). *Mensch und Maschine - Herausforderungen durch Künstliche Intelligenz*. (Stellungnahme). Retrieved from https://www.ethikrat.org/fileadmin/Publikationen/Stellungnahmen/deutsch/stellungnahme-mensch-und-maschine.pdf

Gimpel, H., Graf-Drasch, V., Laubacher, R. J., & Wöhl, M. (2020). Facilitating like darwin: Supporting cross-fertilisation in crowdsourcing. *Decision Support Systems, 132*, 113282. https://doi.org/10.1016/j.dss.2020.113282

Gnewuch, U., Morana, S., & Maedche, A. (2017). Towards designing cooperative and social conversational agents for customer service. *Proceedings of the International Conference on Information Systems (ICIS)*. Seoul, South Korea: Association for Information Systems.

Göldi, A., & Rietsche, R. (2024). Making sense of large language model-based AI agents. *Proceedings of the International Conference on Information Systems (ICIS)*. Bangkok, Thailand: Association for Information Systems. Retrieved from https://aisel.aisnet.org/icis2024/aiinbus/aiinbus/16/

Goodwin, R. (1995). Formalizing properties of agents. *Journal of Logic and Computation, 5*(6), 763–781. https://doi.org/10.1093/logcom/5.6.763

Gregor, S., & Hevner, A.R. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, *37*(2), 337–355. https://doi.org/10.25300/MISQ/2013/37.2.01

Guo, T., Chen, X., Wang, Y., Chang, R., Pei, S., Chawla, N.V., & Zhang, X. (2024). Large language model based multi-agents: A survey of progress and challenges. K. Larson (Ed.), *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI)* (pp. 8048–8057). Jeju, South Korea: International Joint Conferences on Artificial Intelligence Organization.

Gutheil, N., Mayer, V., Müller, L., Römmelt, J., & Kühl, N. (2025). Improving humanAI collaboration through LLM-enhanced prompt engineering. *Proceedings of the International Conference on Information Systems (ICIS)*. Nashville, Tennessee, USA: Association for Information Systems.

Guttman, R. H., Moukas, A. G., & Maes, P. (1998). Agents as meditors in electronic commerce. *Electronic Markets, 8*(1), 22–27. https://doi.org/10.1080/10196789800000007

Haase, J., Kremser, W., Leopold, H., Mendling, J., Onnasch, L., & Plattfaut, R. (2024). Interdisciplinary directions for researching the effects of robotic process automation and large language models on business processes. *Communications of the Association for Information Systems*, *54*(1), 579–604. https://doi.org/10.17705/1CAIS.05421

Händler, T. (2023). *Balancing autonomy and alignment: A multi-dimensional taxonomy for autonomous LLM-powered multi-agent architectures*. arXiv. https://doi.org/10.48550/arXiv.2310.03659

Heerwagen, J. H., Kampschroer, K., Powell, K. M., & Loftness, V. (2004). Collaborative knowledge work environments. *Building Research & Information, 32*(6), 510–528. https://doi.org/10.1080/09613210412331313025

Hemmer, P., Schemmer, M., & Kühl, N., Vössing, M., & Satzger, G. (2025). Complementarity in human-AI collaboration: Concept, sources, and evidence. *European Journal of Information Systems*, *34*(6), 979–1002. https://doi.org/10.1080/0960085X.2025.2475962

Hofmann, P., Samp, C., & Urbach, N. (2020). Robotic process automation. *Electronic Markets, 30*(1), 99–106. https://doi.org/10.1007/s12525-019-00365-8

Hofmann, P., Urbach, N., Lanzl, J., & Desouza, K. C. (2024). AI-enabled information systems: Teaming up with intelligent agents in networked business. *Electronic Markets, 34*, 52. https://doi.org/10.1007/s12525-024-00734-y

Holldack, F., Banh, L., Strobel, G. (2026). Agentic Information Systems. Electronic Markets, 36. https://doi.org/10.1007/12525-025-00861-0

Hübotter, J., Bongni, S., Hakimi, I., Krause, A. (2024). *Efficiently learning at test-time: Active fine-tuning of LLMs*. *Adaptive Foundation Models: Evolving AI for Personalized and Efficient Learning*. Retrieved from https://openreview.net/forum?id=4Tlt7ANaLw

Hughes, L., Dwivedi, Y. K., Malik, T., Shawosh, M., Albashrawi, M. A., Jeon, I., & Walton, P. (2025). AI agents and agentic systems: A multi-expert analysis. *Journal of Computer Information Systems, 65*(4), 489–517. https://doi.org/10.1080/08874417.2025.2483832

IEEE Guide for Terms and Concepts in Intelligent Process Automation. (2017). *IEEE Std 2755-2017* , 1–16. https://doi.org/10.1109/IEEESTD.2017.8070671

Jakubik, J., Vössing, M., Kühl, N., Walk, J., & Satzger, G. (2024). Data-centric artificial intelligence. *Business & Information Systems Engineering, 66*, 507–515. https://doi.org/10.1007/s12599-024-00857-8

Janiesch, C., Zschech, P., & Heinrich, K. (2021). Machine learning and deep learning. *Electronic Markets, 31*(3), 685–695. https://doi.org/10.1007/s12525-021-00475-2

Janz, B., Conquitt, J., & Noe, R. (1997). Knowledge worker team effectiveness: The role of autonomy, interdependence, team development, and contextual support. *Personnel Psychology, 50*(4), 877–904. https://doi.org/10.1111/j.1744-6570.1997.tb01486.x

Jennings, N., & Wooldridge, M. J. (1998). *Agent technology: Foundations, applications, and markets*. New York, NY, USA: Springer Science & Business Media.

Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., & Fung, P. (2023). Survey of hallucination in natural language generation. *ACM Computing Surveys, 55*(12), 1–38. https://doi.org/10.1145/3571730

Jimenez-Ramirez, A., Reijers, H.A., Barba, I., & Del Valle, C. (2019). A method to improve the early stages of the robotic process automation lifecycle. P. Giorgini et al. (Eds.), *Advanced Information Systems Engineering* (Vol. 11483, pp. 446– 461). Cham, Germany: Springer International Publishing.

Kehagias, D. D., Symeonidis, A. L., & Mitkas, P. A. (2005). Designing pricing mechanisms for autonomous agents based on bid-forecasting. *Electronic Markets, 15*(1), 53–62. https://doi.org/10.1080/10196780500035340

Kleinberg, J., & Raghavan, M. (2021). Algorithmic monoculture and social welfare. *Proceedings of the National Academy of Sciences, 118*(22), 1–7. https://doi.org/10.1073/pnas.2018340118

Klesel, M., & Wittmann, H. F. (2025). Retrieval-augmented generation (RAG). *Business & Information Systems Engineering, 67*, 551–561. https://doi.org/10.1007/s12599-025-00945-3

Kuehl, N., Schemmer, M., Goutier, M., & Satzger, G. (2022). Artificial intelligence and machine learning. *Electronic Markets, 32*(4), 2235–2244. https://doi.org/10.1007/s12525-022-00598-0

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Conference on Neural Information Processing Systems* (Vol. 33, pp. 9459–9474). Vancouver, Canada.

Li, G., Hammoud, H.A.A.K., Itani, H., Khizbullin, D., & Ghanem, B. (2023). Camel: Communicative agents for "mind" exploration of large language model society. *Conference on Neural Information Processing Systems* (pp. 51991–52008). New Orleans, Louisiana, USA.

Li, X., Wang, S., Zeng, S., Wu, Y., & Yang, Y. (2024). A survey on LLM-based multiagent systems: Workflow, infrastructure, and challenges. *Vicinagearth, 1*, 9. https://doi.org/10.1007/s44336-024-00009-2

Li, X., Shang, W., Niu, Z., Xie, B., Zhu, L. (2025). Can large language models evaluate suppliers' qualifications of complex production lines? An approach incorporating retrieval-augmented generation

and chain-of-thought. *Proceedings of the International Conference on Information Systems (ICIS)*. Nashville, Tennessee, USA: Association for Information Systems.

Li, Y., Sun, L., & Zhang, Y. (2025). MetaAgents: Large language model based agents for decision-making on teaming. *Proceedings of the ACM on Human Computer Interaction, 9*(2), 1–27. https://doi.org/10.1145/3711032

Liang, Y., Wu, C., Song, T., Wu, W., Xia, Y., Liu, Y., Ou, Y., Lu, S., Ji, L., Mao, S., Wang, Y., Shou, L., Gong, M., & Duan, N. (2024). TaskMatrix.AI: Completing tasks by connecting foundation models with millions of APIs. *Intelligent Computing*, *3* , 0063. https://doi.org/10.34133/icomputing.0063

Lins, S., Pandl, K. D., Teigeler, H., Thiebes, S., Bayer, C., & Sunyaev, A. (2021). Artificial intelligence as a service. *Business & Information Systems Engineering, 63*(4), 441–456. https://doi.org/10.1007/s12599-021-00708-w

Liu, P., Yuan, W., Fu, J., Jiang, Z., Hayashi, H., & Neubig, G. (2023). Pre-train, prompt, and predict: A systematic survey of prompting methods in natural language processing. *ACM Computing Surveys, 55*(9), 195. https://doi.org/10.1145/3560815

Liu, Y., Lo, S. K., Lu, Q., Zhu, L., Zhao, D., Xu, X., Harrer, S, & Whittle, J. (2025). Agent design pattern catalogue: A collection of architectural patterns for foundation model based agents. *Journal of Systems and Software, 220*, 112278. https://doi.org/10.1016/j.jss.2024.112278

Lu, J., Pan, B., Chen, J., Feng, Y., Hu, J., Peng, Y., & Chen, W. (2024). Agentlens: Visual analysis for agent behaviors in LLM-based autonomous systems. *IEEE Transactions on Visualization and Computer Graphics, 31*(8), 4182–4197. https://doi.org/10.1109/TVCG.2024.3394053

Lu, P., Peng, B., Cheng, H., Galley, M., Chang, K.W., Wu, Y.N., Zhu, S.C., & Gao, J. (2023). Chameleon: Plug-and-play compositional reasoning with large language models. *Conference on Neural Information Processing Systems* (Vol. 36, pp. 43447–43478). New Orleans, Louisiana, USA: Curran Associates, Inc.

Mandi, Z., Jain, S., Song, S. (2024). RoCo: Dialectic multi-robot collaboration with large language models. *2024 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 286–299). IEEE.

Mao, S., Cai, Y., Xia, Y., Wu, W., Wang, X., Wang, F., Ge, T., & Wei, F. (2023). *ALYMPICS: LLM agents meet game theory – exploring strategic decisionmaking with AI agents*. arXiv. https://doi.org/10.48550/arXiv.2311.03220

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., Zettlemoyer, L. (2022). Rethinking the role of demonstrations: What makes in-context learning work? Y. Goldberg et al. (Eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 11048–11064). Stroudsburg, PA, USA: Association for Computational Linguistics.

Mu, C., Guo, H., Chen, Y., Shen, C., Hu, D., Hu, S., & Wang, Z. (2024). Multi-agent, human-agent and beyond: A survey on cooperation in social dilemmas. *Neurocomputing, 610*, 128514. https://doi.org/10.1016/j.neucom.2024.128514

Nambisan, S. (2000). EC and supply chain management: Towards cross-industry supply chains. *Electronic Markets, 10*(3), 197–202. https://doi.org/10.1080/10196780050177116

O'Brien, P. D., & Nicol, R. C. (1998). FIPA-towards a standard for software agents. *BT Technology Journal, 16*(3), 51–59. https://doi.org/10.1023/A:1009621729979

Oliver, J. R. (1996). A machine-learning approach to automated negotiation and prospects for electronic commerce. *Journal of Management Information Systems, 13*(3), 83–112. https://doi.org/10.1080/07421222.1996.11518135

Pan, M. Z., Cemri, M., Yang, S., Agrawal, L.A., Chopra, B., Tiwari, R., Keutzer, K., Parameswaran, A., Klein, D., Ramchandran, K., Zaharia, M., Gonzalez, J.E., & Stoica, I. (2025). Why do multiagent systems fail? *ICLR 2025 Workshop on Building Trust in Language Models and Applications*.

Peffers, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems, 24*(3), 45–77. https://doi.org/10.2753/MIS0742-1222240302

Pyöriä, P. (2005). The concept of knowledge work revisited. *Journal of Knowledge Management, 9*(3), 116–127. https://doi.org/10.1108/13673270510602818

Rafailov, R., Sharma, A., Mitchell, E., Manning, C.D., Ermon, S., Finn, C. (2023). Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems* (Vol. 36, pp. 53728–53741). New Orleans, Louisiana, USA: Curran Associates, Inc.

Ray, P. P. (2023). ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope. *Internet of Things and Cyber-Physical Systems, 3*, 121–154. https://doi.org/10.1016/j.iotcps.2023.04.003

Russell, S., Norvig, P., Intelligence, A. (1995). *Artificial intelligence: A modern approach* (First ed.). Pearson Education Limited.

Sapkota, R., Roumeliotis, K.I., Karkee, M. (2026). AI agents vs. agentic AI: A conceptual taxonomy, applications and challenges. *Information Fusion*, *126*, 103599. https://doi.org/10.1016/j.inffus.2025.103599

Schemmer, M., Kuehl, N., Benz, C., Bartos, A., Satzger, G. (2023). Appropriate reliance on ai advice: Conceptualization and the effect of explanations. *International Conference on Intelligent User Interfaces* (pp. 410–422). Sydney, NSW, Australia: Association for Computing Machinery.

Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Zettlemoyer, L., Cancedda, N., & Scialom, T. (2023). Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems* (Vol. 36, pp. 68539–68551). Curran Associates, Inc.

Schmidt, R., Alt, R., & Zimmermann, A. (2023). Assistant platforms. *Electronic Markets, 33*(1), 59. https://doi.org/10.1007/s12525-023-00671-2

Schneider, J., Meske, C., & Kuss, P. (2024). Foundation models: A new paradigm for artificial intelligence. *Business & Information Systems Engineering, 66*(2), 221–231. https://doi.org/10.1007/s12599-024-00851-0

Schuetz, S., & Venkatesh, V. (2020). The rise of human machines: How cognitive computing systems challenge assumptions of user-system interaction. *Journal of the Association for Information Systems*, *21*(2), 460–482

Schultze, U. (2000). A confessional account of an ethnography about knowledge work. *MIS Quarterly, 24*(1), 3–41. https://doi.org/10.2307/3250978

Shen, Y., Song, K., Tan, X., Li, D., Lu, W., Zhuang, Y. (2023). HuggingGPT: Solving AI tasks with ChatGPT and its friends in hugging face. *Advances in Neural Information Processing Systems* (Vol. 36, pp. 38154–38180). New Orleans, Louisiana, USA: Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paperfiles/paper/2023/file/77c33e6a367922d003ff102ffb92b658-Paper-Conference.pdf

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. A. Oh et al. (Eds.), *Advances in Neural Information Processing Systems* (Vol. 36, pp. 8634–8652). New Orleans, Louisiana, USA: Curran Associates, Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paperfiles/paper/2023/file/1b44b878bb782e6954cd888628510e90-Paper-Conference.pdf

Singh, R., Salam, A. F., & Iyer, L. (2005). Agents in e-supply chains. *Communications of the ACM, 48*(6), 108–115. https://doi.org/10.1145/1064830.1064835

Smith, R. G. (1980). The contract net protocol: High-level communication and control in a distributed problem solver. *IEEE Transactions on Computers, 29*(12), 1104–1113. https://doi.org/10.1109/TC.1980.1675516

Song, T., Tan, Y., Zhu, Z., Feng, Y., Lee, Y.-C. (2025). Multiagents are social groups: Investigating social influence of multiple agents in human-agent interactions. *Proceedings of the ACM on Human Computer Interaction*, *9*(7), 1–33). https://doi.org/10.1145/3757633

Sterz, S., Baum, K., Biewer, S., Hermanns, H., Lauber-Rönsberg, A., Meinel, P., Langer, M. (2024). On the quest for effectiveness in human oversight: Interdisciplinary perspectives. *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (pp. 2495–2507). Rio de Janeiro, Brazil: ACM.

Surapaneni, R., Jha, M., Vakoc, M., Segal, T. (2025). *Announcing the agent2agent protocol (a2a)*. Retrieved 2025-09-13, from https://developers.googleblog.com/ en/a2a-a-new-era-of-agent-interoperability/

Svanberg, M., Li, W., Fleming, M., Goehring, B., Thompson, N. (2024). Beyond AI exposure: Which tasks are cost-effective to automate with computer vision? *Available at SSRN 4700751*, 1–45, Retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstractid=4700751

Tapia, D.I., Alonso, R.S., Zato, C., Gil, O., de La Prieta, F. (2010). Analysis and design of a SOA-based multi-agent architecture. *Trends in Practical Applications of Agents and Multiagent Systems* (Vol. 71, pp. 183–190). Berlin, Germany: Springer.

Tuunanen, T., Winter, R., vom Brocke, J. (2024). Dealing with complexity in design science research: A methodology using design echelons. *MIS Quarterly*, *48*(2), 427–458. https://doi.org/10.25300/MISQ/2023/16700

Van der Aalst, W., ter Hofstede, A., Kiepuszewski, B., & Barros, A. P. (2003). Workflow patterns. *Distributed and Parallel Databases, 14*(1), 5–51. https://doi.org/10.1023/A:1022883727209

Vernon, D., Metta, G., & Sandini, G. (2007). A survey of artificial cognitive systems: Implications for the autonomous development of mental capabilities in computational agents. *IEEE Transactions on Evolutionary Computation, 11*(2), 151–180. https://doi.org/10.1109/TEVC.2006.890274

Wang, L., Ma, C., Feng, X., Zhang, Z., Yang, H., Zhang, J., & Wen, J. (2024). A survey on large language model based autonomous agents. *Frontiers of Computer Science, 18*, 186345. https://doi.org/10.1007/s11704-024-40231-1

Webb, T., Holyoak, K. J., & Lu, H. (2023). Emergent analogical reasoning in large language models. *Nature Human Behaviour,7*(9), 1526–1541. https://doi.org/10.1038/s41562-023-01659-w

Wei, A., Haghtalab, N., Steinhardt, J. (2023). Jailbroken: How does LLM safety training fail? *Advances in Neural Information Processing Systems* (Vol. 36, pp. 80079–80110). Curran Associates, Inc.

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems* (Vol. 35, pp. 24824–24837). Curran Associates, Inc.

Willcocks, L., Lacity, M., & Craig, A. (2017). Robotic process automation: Strategic transformation lever for global business services? *Journal of Information Technology Teaching Cases, 7*(1), 17–28. https://doi.org/10.1057/s41266-016-0016-9

Wooldridge, M. (2009). *An introduction to multiagent systems*. John Wiley & Sons.

Wooldridge, M., & Jennings, N. R. (1995). Intelligent agents: Theory and practice. *The Knowledge Engineering Review, 10*(2), 115–152. https://doi.org/10.1017/S0269888900008122

Wu, D., Sun, Y., & Zhong, F. (2000). Organizational agent systems for intelligent enterprise modelling. *Electronic Markets, 10*(4), 272–281. https://doi.org/10.1080/10196780010001303

Wu, D., Lin, J., Li, Z.A. (2025). Foundation models and AI innovation: Evidence from the hugging face platform.

Xi, Z., Chen, W., Guo, X., He, W., Ding, Y., Hong, B., & Gui, T. (2025). The rise and potential of large language model based agents: A survey. *Science China Information Sciences*, *68*(2), 121101. https://doi.org/10.1007/s11432-024-4222-0

Xiao, B., & Benbasat, I. (2007). E-commerce product recommendation agents: Use, characteristics, and impact. *MIS Quarterly, 137–209*. https://doi.org/10.2307/25148784

Yang, C., Pereira Nunes, B., Rodríguez Méndez, S. (2025). LLM as auto-prompt engineer: Automated NER prompt optimisation. *Companion Proceedings of the ACM on Web Conference 2025* (pp. 2574–2578). New York, NY, USA: Association for Computing Machinery.

Zhou, J., Lu, Q., Chen, J., Zhu, L., Xu, X., Xing, Z., Harrer, S. (2024). *A taxonomy of architecture options for foundation model-based agents: Analysis and decision model*. arXiv. https://doi.org/10.48550/arXiv.2408.02920

Zhou, W., Jiang, Y.E., Li, L., Wu, J., Wang, T., Qiu, S., Zhang, J., Chen, J., Wu, R., Wang, S., Zhu, S., Chen, J., Zhang, W., Tang, X., Zhang, N., Chen, H., Cui, P., & Sachan, M. (2024). Agents: An open-source framework for autonomous language agents. *ICLR 2024 Workshop on Large Language Model (LLM) Agents*. Retrieved from https://openreview.net/forum?id=wEPsLQ4QpM

Zhu, X.X., Xiong, Z., Wang, Y., Stewart, A.J., Heidler, K., Wang, Y., Yuan, Z., Dujardin, T., Xu, Q., & Shi, Y. (2024). *On the foundations of earth and climate foundation models*. arXiv. https://doi.org/10.48550/arXiv.2405.04285

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